

CALIBRATION-RISK ROUTING FOR CONTROLLED WORLD-MODEL ADAPTATION

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ABSTRACT

Model-based reinforcement learning (MBRL) can exploit simulated experience, but a simulator-to-target shift creates a model-selection problem: correcting the simulator and fitting the target directly can each fail under limited target data. We introduce the Model-Corrected World Model (MC-WM), which separates initial target data into disjoint fit, selection, and calibration partitions and deploys the family with lower standardized calibration risk. A learned confidence signal and deterministic validity predicates weight one-step imagined policy updates without rewriting physical rewards. We evaluate 540 unique reported run cells across three controlled Multi-Joint dynamics with Contact (MuJoCo) shifts; one exact-routing cell was repeated after a pre-deployment artifact gate, giving 541 completed executions. In a fresh ten-seed confirmation, MC-WM exceeds target Soft Actor-Critic (SAC) return by 1925.7 [715.6, 3322.3], 1184.5 [724.5, 1752.7], and 133.4 [67.6, 197.9] on Ant actuator, HalfCheetah gravity, and Walker friction, respectively, while reducing violations in all three. It also exceeds matched Domain Adaptation with Rewards from Classifiers (DARC) return in every environment, but its return differences from direct target Model-Based Policy Optimization (MBPO) remain unresolved. The selector chooses direct target models for all Ant and gravity runs and residual models for all Walker runs. Automatic routing exactly reproduces its selected forced arm in all 30 paired checks and improves Ant return over the unselected arm by 1529.9 [426.9, 2833.1]; gravity and Walker return and all route-level violation differences remain unresolved. A separate sparse-versus-dense study finds fewer residual parameters but no resolved endpoint advantage. The evidence supports calibrated model-family choice and bounded trust mechanisms, not universal residual or symbolic superiority.

1 INTRODUCTION

Simulation can supply control data without exposing a target system to every exploratory action. Dynamics randomization and rapid adaptation use this advantage to train reinforcement learning (RL) policies that tolerate or identify dynamics variation (Peng et al., 2017; Kumar et al., 2021). MBRL instead fits a predictive model and improves the policy with imagined transitions. Under a simulator-to-target dynamics shift, however, either reusing the simulator model or fitting a target model from limited data can fail. Policy optimization can then exploit errors that were rare under the model-fitting distribution.

Probabilistic ensembles represent model uncertainty (Chua et al., 2018), and short branched rollouts limit compounding error (Janner et al., 2019), but neither decides whether simulator correction or direct target modeling is better for a particular shift. Model fitting must also account for the state-action distribution induced by policy learning (Ross & Bagnell, 2012). This leaves two coupled questions: which target-model family should be deployed, and how strongly should each imagined transition influence policy learning?

We study these questions in MC-WM. MC-WM fits two candidates from the same initial target data: a residual correction to a frozen simulator ensemble and a direct target ensemble. A disjoint calibration partition measures standardized joint state-and-reward error and fixes the lower-risk family

for the run. Direct-candidate fitting uses an isolated random stream, so model-family selection cannot change downstream policy or replay initialization merely by consuming random numbers. The deployed model then produces one-step imagined transitions whose policy-critic losses are weighted by a learned confidence signal and deterministic validity predicates. On the residual route, the state correction combines a locked sparse component with a masked neural remainder; the support, coefficients, calibration errors, routing decision, filter behavior, and target access are exported for every run.

The paper makes three contributions. First, it defines calibration-risk routing between residual and direct target world models using disjoint fit, selection, and calibration partitions. Second, it provides an exact routing counterfactual: automatic, forced-residual, and forced-direct arms fit identical candidates under matched random streams, and the automatic arm must exactly reproduce its selected forced arm. Third, it evaluates the complete method and its limits through 540 unique reported run cells from 541 completed executions: a fresh ten-seed main comparison on three controlled dynamics shifts and morphologies, the routing counterfactual, a source-identical predecessor mechanism matrix, and a matched sparse-versus-dense residual study. The evidence is deliberately bounded to target MuJoCo simulators. It does not imply physical sim-to-real transfer, stable symbolic discovery across seeds, reduced total target access, or a safety guarantee.

2 RELATED WORK

Adaptation across dynamics. Visual domain randomization trains perception networks across randomized rendering conditions (Tobin et al., 2017); dynamics randomization trains control policies across randomized physics (Peng et al., 2017). Rapid Motor Adaptation (RMA) infers an environment representation online (Kumar et al., 2021). DARC estimates a source-to-target reward correction from state-action and transition domain classifiers (Eysenbach et al., 2020). Hybrid Offline-and-Online RL with dynamics gaps (H2O+) combines limited offline target data with online simulator interaction (Niu et al., 2023). MC-WM differs by fitting explicit residual and direct target world-model candidates, selecting between them on held-out transition risk, and weighting model-generated policy updates. DARC is included under the same target-training and held-out evaluation contract as the other main methods.

Residual world-model adaptation is the closest model class. Lanier et al. (2025) adapt visual latent world models using offline, reward-unlabelled target data and latent-state dynamics residuals. MC-WM uses reward-labelled online target data and observation-space increments, places a locked sparse component and masked neural remainder on the residual route, and allows calibration risk to reject that route in favor of a direct target ensemble. These are architectural and protocol differences; they do not establish dominance across the distinct visual-control and state-control settings.

World models and policy optimization. Probabilistic Ensembles with Trajectory Sampling (PETS) represents predictive uncertainty with probabilistic dynamics ensembles (Chua et al., 2018). MBPO limits model bias with short rollouts branched from real data (Janner et al., 2019). Dreamer learns behaviors through latent imagination (Hafner et al., 2019). Temporal Difference Learning for Model Predictive Control (TD-MPC) combines latent dynamics with a terminal value estimate for planning (Hansen et al., 2022). MC-WM uses one-step rollouts so that the experiment primarily tests target-model choice, correction, and weighting rather than long-horizon compounding.

Structured residuals and constrained updates. Sparse Identification of Nonlinear Dynamics (SINDy) estimates sparse dynamics from a prescribed feature library (Brunton et al., 2015). MC-WM applies its sparse-regression principle to target-minus-simulator state increments, but the sparse branch is evaluated as a capacity and inspectability choice rather than assumed to improve control. Policy learning is based on SAC (Haarnoja et al., 2018). The project-specific three-critic pessimistic actor is related to ensemble methods such as Randomized Ensembled Double Q-Learning (REDQ), but is not a REDQ implementation (Chen et al., 2021). Constrained Policy Optimization optimizes an expected-cost constraint (Achiam et al., 2017), whereas Inverse Constrained Reinforcement Learning infers constraints from demonstrations (Anwar et al., 2020). MC-WM’s predicates are narrower: they screen imagined transitions and do not constrain the executed target policy.

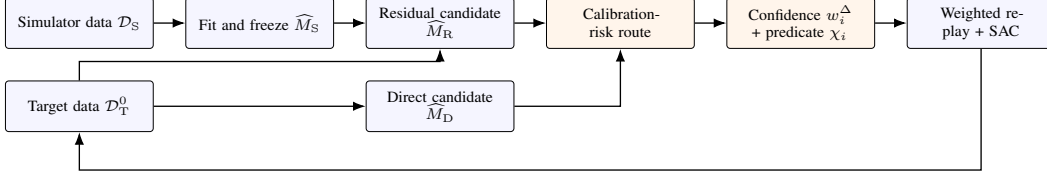


Figure 1: MC-WM execution path. Initial target data fit a residual and a direct candidate; a disjoint calibration partition fixes the deployed family. Online target replay refits only the deployed state-and-reward model. Trust weights scale losses rather than physical rewards; a separate violation cost is described in Section 3.5.

3 METHOD

3.1 PROBLEM SETTING

We use observed-state Markov decision process (MDP) approximations for the simulator and target, $\mathcal{M}_\ell = (\mathcal{S}, \mathcal{A}, P_\ell, \rho_\ell, \gamma, p_0)$ indexed by $\ell \in \{S, T\}$. They share the state space $\mathcal{S}$, action space $\mathcal{A}$, policy-return discount $\gamma \in [0, 1)$, and initial-state distribution p_0 , but may differ in the transition kernel P_ℓ and expected one-step reward ρ_ℓ . The target objective is

$$J_{\mathcal{M}_T}(\pi_\phi) = \mathbb{E}_{\mathcal{M}_T, \pi_\phi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right], \quad (1)$$

where π_ϕ is the policy with parameters ϕ and r_t is the observed target reward, with zero rewards after episode termination. Equation (1) defines the discounted task objective; training uses the entropy-regularized, violation-aware surrogate in Section 3.5, and evaluation reports undiscounted episode return. Observed state, action, and termination at time t are $s_t \in \mathbb{R}^{n_s}$, $a_t \in \mathbb{R}^{n_a}$, and $\kappa_t \in \{0, 1\}$; $x_t = [s_t; a_t]$ is the state-action input. The simulator dataset is $\mathcal{D}_S$, and the initial target dataset is $\mathcal{D}_T^0$. Appendix A gives the complete one-symbol-one-meaning ledger.

3.2 PIPELINE AND CALIBRATION-RISK ROUTING

MC-WM fits two target-model candidates before online policy learning (Figure 1; Algorithm 1 in Appendix E.2). The residual candidate $\widehat{M}_R$ corrects a frozen simulator ensemble $\widehat{M}_S$, whereas the direct candidate $\widehat{M}_D$ fits the same probabilistic ensemble architecture directly to target transitions. An untouched calibration split selects the candidate with lower standardized one-step state-and-reward error. This route is fixed for the run. The deployed model then supplies confidence-weighted one-step transitions for target-domain SAC updates.

The initial 10,000 target transitions are split reproducibly into index sets $\mathcal{I}_{\text{fit}}$, $\mathcal{I}_{\text{sel}}$, and $\mathcal{I}_{\text{cal}}$ containing 80%, 10%, and 10% of the rows. The fit split estimates parameters; the selection split chooses sparse features and supplies early stopping for the direct ensemble; neither candidate uses the calibration split during fitting or early stopping. Define the joint one-step target

$$z_i = [s_{i+1} - s_i; r_i] \in \mathbb{R}^{n_s+1}. \quad (2)$$

For a dataset sample i , s_{i+1} denotes its observed successor, not the next row of a shuffled dataset. For candidate $m \in \{R, D\}$, let $\widehat{z}_i^{(m)}$ be its deterministic ensemble-mean prediction and let ς_k be the standard deviation of coordinate k of z_i on $\mathcal{I}_{\text{fit}}$, replaced by one when it is at most 10^{-6} . The registered routing risk is

$$\mathcal{R}_m = \left[\frac{1}{|\mathcal{I}_{\text{cal}}|(n_s+1)} \sum_{i \in \mathcal{I}_{\text{cal}}} \sum_{k=1}^{n_s+1} \left(\frac{\widehat{z}_{i,k}^{(m)} - z_{i,k}}{\varsigma_k} \right)^2 \right]^{1/2}. \quad (3)$$

Equation (3) evaluates the target in Equation (2) after coordinate standardization. MC-WM deploys

$$m^* = \begin{cases} D, & \mathcal{R}_D < \mathcal{R}_R, \\ R, & \text{otherwise,} \end{cases} \quad \widehat{M}_T = \widehat{M}_{m^*}. \quad (4)$$

Thus a tie retains the residual candidate. After Equation (4) fixes the route, all initial target rows enter online replay. No validation episode or test trajectory participates in this model-family decision.

Each probabilistic ensemble contains $K = 5$ multilayer perceptron (MLP) members. Every MLP has three 200-unit sigmoid linear unit (SiLU) hidden layers and predicts a Gaussian mean and diagonal variance for state increment and reward. Input and target normalizers are fixed at the candidate’s initial fit. On the direct route, $\widehat{M}_D$ is warm-start refit every 1,000 online target steps on at most 50,000 replay transitions. On the residual route, the correction described next is refit at the same frequency and replay cap while $\widehat{M}_S$ remains frozen.

Direct-candidate fitting uses a dedicated random stream derived from the seed, preserving subsequent policy, replay, and optimizer initialization. Automatic and forced-route arms fit both candidates in the same order and intervene only on deployment in Equation (4). The automatic arm must exactly reproduce its selected forced arm; Appendix C.1 details this isolation and matching test.

3.3 LOCKED-SUPPORT RESIDUAL CANDIDATE

Let $\widehat{u}_S(x)$ and $\widehat{\rho}_S(x)$ denote the frozen simulator ensemble’s mean state-increment and reward predictions. For target transition i , the state correction target is

$$y_i^{\Delta s} = s_{i+1} - s_i - \widehat{u}_S(x_i). \quad (5)$$

Equation (5) removes the simulator-predicted increment from the observed target increment. The residual candidate predicts

$$\widehat{s}_{t+1}^{(R)} = s_t + \widehat{u}_S(x_t) + \widehat{\Delta}_s(x_t), \quad (6)$$

$$\widehat{r}_t^{(R)} = \widehat{\rho}_S(x_t) + \widehat{\Delta}_{r,\zeta}(x_t), \quad (7)$$

Equations (6) and (7) define the residual candidate’s state and reward predictions, respectively, where $\widehat{\Delta}_{r,\zeta}$ is a one-hidden-layer, 64-unit rectified linear unit (ReLU) reward correction with parameters ζ . The additive state correction is

$$\widehat{\Delta}_s(x) = B\widetilde{\Theta}(x) + g_\psi(\mu \odot \widetilde{\Theta}(x)). \quad (8)$$

Equation (8) defines the complete state correction. Here, $\mathcal{F}$ is the candidate feature set with a fixed evaluation order, $\Theta(x)$ evaluates its features as a vector, $\widetilde{\Theta}(x) \in \mathbb{R}^p$ is its root-mean-square-scaled deployed subset, $B \in \mathbb{R}^{n_s \times p}$ is the sparse coefficient matrix, $\mu \in \{0, 1\}^p$ marks features active in at least one state coordinate, g_ψ is the masked neural remainder, and $\odot$ is elementwise multiplication. The remainder has one 32-unit SiLU hidden layer and a zero-initialized output layer. It therefore starts at zero and cannot consume a feature outside the reported support.

A degree-two polynomial library supplies the base features. Sequentially thresholded least squares (STLSQ) uses fixed ridge and threshold values; deterministic additions must pass a selection-partition mean squared error (MSE) gate and a coefficient screen. The accepted feature order and support are then locked. Online residual refits update coefficients, the masked neural remainder, reward, and eligible termination heads without adding features, and weight rows by proximity to recent policy states. Appendix E.1 gives the selection gate in Equation (17), the refit weight in Equation (18), and the termination prediction in Equation (19).

3.4 CONFIDENCE AND PREDICATE WEIGHTS

For target sample i , define the deployed-model state error $e_i = n_s^{-1} \|\widehat{s}_{i+1} - s_{i+1}\|_2^2$. Let τ_e be an positive exponential moving average of the batch median error, initialized at 10^{-3} and updated with weight 0.1 on the new median (floored at 10^{-8}), and $c_i = (1 + e_i/\tau_e)^{-1}$. MC-WM fits a nonpositive log-confidence critic Q_ξ^Δ . Its ensemble signature $\sigma_{\text{ens}}(s, a) \in \mathbb{R}_{\geq 0}^{n_s+1}$ is the coordinatewise standard deviation of the direct target ensemble on the direct route and of the frozen simulator ensemble on the residual route. The regression target is

$$q_i^{\text{tgt}} = \log \max(c_i, 10^{-8}) + \gamma_\Delta (1 - \kappa_i) Q_\xi^\Delta(s_{i+1}, a_{i+1}^{\text{det}}, \sigma_{\text{ens}}(s_{i+1}, a_{i+1}^{\text{det}})), \quad (9)$$

where a_{i+1}^{det} is the current policy’s deterministic action (the squashed Gaussian mean) at s_{i+1} , ξ is the confidence-critic parameter vector, $\bar{\xi}$ denotes its Polyak-averaged target parameters, and $\gamma_\Delta = 0.5$ is distinct from the policy discount. The imagined transition weight is

$$w_i^\Delta = \exp(Q_\xi^\Delta(s_i, a_i, \sigma_{\text{ens}}(s_i, a_i))) \in (0, 1]. \quad (10)$$

Equation (9) trains the critic, while Equation (10) converts its nonpositive output to a weight. This value is a learned ranking signal, not a calibrated probability.

The deterministic predicate set $\mathcal{C}$ contains broad morphology-specific representation, transition-magnitude, and reward-range checks. The Walker set additionally rejects non-finite predictions. The HalfCheetah and Walker experiments use nonempty sets; the Ant environment uses an empty set. For imagined transition i , the indicator χ_i is one only when every configured predicate passes. The final trust weight is

$$w_i = w_i^\Delta \chi_i, \quad (11)$$

and rows with $w_i \leq 0.01$ are discarded. Predicates remove known invalid predictions but do not constrain target actions.

3.5 VIOLATION-AWARE POLICY OPTIMIZATION AND CHECKPOINT SELECTION

Every 250 target steps, the current policy acts from 400 target-replay states, and $\widehat{M}_T$ generates one-step transitions. Physical rewards are stored without confidence scaling. A separate binary violation indicator $\iota_i \in \{0, 1\}$ is supplied by target-environment diagnostics for real transitions and by the same environment’s state predicate for imagined transitions. At online step $t \in \{1, \dots, N_{\text{on}}\}$, replay sampling uses

$$\lambda_v(t) = 10 \max\left(0, \frac{t - 0.5N_{\text{on}}}{0.5N_{\text{on}}}\right), \quad r_i^{\text{train}} = r_i - \lambda_v(t)\iota_i. \quad (12)$$

Thus Equation (12) leaves the violation cost at zero through the first half of online training and raises it linearly to 10; the cost is distinct from the trust weight in Equation (11).

Let $\mathcal{I}$ index a minibatch of size $b = 256$, let Q_{η_j} be policy critic $j \in \{1, 2, 3\}$ with parameters η_j , and let y_i^Q be the SAC target formed with scale-normalized r_i^{train} , the minimum of the three target critics, and the entropy-adjusted stochastic next action. The critic loss is

$$\mathcal{L}_Q = \frac{1}{3b} \sum_{j=1}^3 \sum_{i \in \mathcal{I}} w_i \left(Q_{\eta_j}(s_i, a_i) - y_i^Q \right)^2 + \lambda_{\text{ens}} \frac{1}{b} \sum_{i \in \mathcal{I}} \text{Std}_j [Q_{\eta_j}(s_i, a_i)], \quad (13)$$

In Equation (13), $w_i = 1$ for target transitions, $\lambda_{\text{ens}} = 0.01$, and Std_j takes the sample standard deviation across the three policy critics. The actor maximizes the critic mean minus two critic standard deviations plus the SAC entropy term, with a 10^{-3} behavior-cloning MSE penalty. Policy learners use scale-only running reward normalization.

An initial simulator-to-residual state root mean squared error (RMSE) ratio ϱ_{cal} on $\mathcal{I}_{\text{cal}}$ fixes whether actor updates mix target and imagined states; this actor route is separate from Equation (4). Fixed validation checkpoints may restore an eligible actor before the one-time held-out test, so validation is counted as model-selection access. The exact actor band, restoration inequalities, and complete training procedure appear in Appendix E.2.

4 EXPERIMENTAL PROTOCOL

4.1 CONTROLLED TARGET DOMAINS

Experiments use the MuJoCo physics engine (Todorov et al., 2012) through Gymnasium 1.2.0 v5 tasks. Each simulator–target pair preserves the task morphology and observation and action dimensions while changing one controlled dynamics factor. The final-confirmation matrix contains three environments, spanning three morphologies and three shift types (Table 1). Appendix F gives the exact physics, reward, and termination transformations. We call the held-out member of each pair the *target*; these are simulator-to-simulator experiments, not physical sim-to-real deployments.

The task coordinates are q_z , HalfCheetah’s root vertical generalized coordinate at observation index 0; v_x , Ant’s forward velocity at index 13; and $s^{(0)}$, Walker’s root-height observation. The HalfCheetah coordinate is zero-centered, not absolute torso height. Gravity is in m s^{-2} .

Table 1: Controlled simulator-to-target environments. Each reported constraint is identical in the simulator and target member of a pair.

Name	Base task	Simulator $\rightarrow$ target dynamics	Counted violation
Ant actuator	Ant-v5	8 active $\rightarrow$ actions 4–7 disabled	$ v_x > 1.5$ or rear-wall hit
Gravity	HalfCheetah-v5	vertical gravity $-19.62 \rightarrow -9.81$	$q_z > 0.2$
Friction Walker	Walker2d-v5	friction factor $0.3 \rightarrow 1.0$	$s^{(0)} > 1.25$

The Gravity and Friction Walker thresholds apply continuous quadratic reward penalties, whereas the Ant environment combines a quadratic velocity penalty with termination when the world-frame base position is below -3 along the first horizontal coordinate. A violation is counted at each penalized step and at a wall termination. Counts are therefore comparable only within an environment and are never pooled across constraint definitions.

The Ant construction is inspired by the Ant-Broken constraint-transfer setting of [Anwar et al. \(2020\)](#), but it is not a reproduction. It uses a 27-dimensional Gymnasium-v5 observation, four disabled actuator channels, a project-specific radial reward, and a shared velocity cap. The deterministic predicate set used by MC-WM does not contain an Ant-specific predicate in the reported protocol; Ant violations remain environment outcomes and training costs rather than predicate-filter events.

4.2 CANDIDATE FREEZING AND MAIN COMPARATORS

The final candidate was fixed before the confirmation endpoints were read. A registered seven-seed development screen replaced the predecessor’s indirect state-error-gap route with the calibration risk in Equation (3). A subsequent implementation audit found that fitting the direct candidate had consumed the global random stream before policy initialization. We corrected this causal confound by isolating the component random stream as described in Section 3.2, froze the corrected source, and discarded the affected confirmation endpoints from the submission evidence. The final ten-seed bank was then sampled separately. It excludes all 40 seeds registered by the preceding development and confirmation studies. Development outcomes are used only for algorithm selection; all main performance claims use the fresh, random-stream-isolated confirmation matrix.

The main matrix contains the frozen calibration-routed MC-WM candidate in Algorithm 1 and five controls. *Residual MC-WM core* retains the sparse-plus-neural residual, confidence critic, and policy-aware refitting, but fixes residual deployment and removes adaptive update routing, the violation cost, and validation-based checkpoint restoration. *Direct target MBPO* fits the same probabilistic world-model architecture directly to target transitions and uses one-step rollouts. *Target SAC* learns from target transitions without a learned world model. Our matched-protocol *DARC* implementation uses the published additive state-action and state-action-next-state classifiers, 32-unit hidden layers, unit Gaussian input noise, and starts reward correction after 4,000 online updates ([Eysenbach et al., 2020](#)). The warmup is scaled to our 40,000-step online phase and is not presented as a reproduction of DARC’s original longer schedule. *Raw simulator SAC* learns for 50,000 interactions in the uncorrected simulator and receives no target-domain training transition. All SAC-labelled methods use the same project-specific policy backbone described in Section 3.5; “SAC” does not denote an otherwise unchanged reference implementation.

The primary comparisons are candidate versus residual MC-WM core and direct target MBPO. They test whether calibrated family selection improves over a fixed residual and whether simulator correction improves over direct target modeling. Target SAC, DARC, and raw simulator SAC are secondary controls for model-free target learning, classifier-based adaptation, and zero-target-training transfer, respectively. Every method is run for ten independently trained seeds on all three environments, giving $3 \times 6 \times 10 = 180$ confirmation runs from one frozen source tree.

4.3 TRAINING AND EVALUATION ACCESS

Table 2 separates interactions that can change model or policy parameters from target validation and final testing. Each adaptive method receives 10,000 initial target transitions and 40,000 online target transitions. Candidate and residual runs also use 50,000 simulator transitions to fit the frozen source model. DARC uses 10,000 initial and 40,000 online simulator transitions. Direct target MBPO and target SAC use no simulator training transition. Raw simulator SAC instead uses 50,000 simulator transitions and zero target training transitions.

Whenever a method receives an initial simulator or target dataset, each action coordinate is sampled independently and uniformly within the environment’s action bounds using a dataset-specific seed. Online loops index interactions from $t = 1$: actions are sampled from the same uniform distribution for $t < 2,000$ (steps 1–1,999), and from the stochastic policy beginning at $t = 2,000$. DARC applies this online cutoff simultaneously in its simulator and target streams; raw simulator SAC applies it only in the simulator. These random-action transitions are included in the training budgets in Table 2.

Validation uses seeds 10,000–10,004 for five episodes at each of eight checkpoints. Adaptive methods are evaluated every 5,000 online steps; raw simulator SAC is evaluated every 6,250 simulator steps so that it has the same number of validation calls over its 50,000-step budget. The one-shot final test uses seeds 20,000–20,019 for 20 episodes. An episode has at most 1,000 steps, so an adaptive run can access at most 110,000 target transitions: 50,000 for training, 40,000 for validation, and 20,000 for final testing. Early terminations reduce the realized count.

Validation data never enter gradient updates. They are nevertheless model- selection access: the candidate and its exact routing variants can restore a registered actor checkpoint from the validation trajectory. We therefore report training, validation, test, and total target access separately and do not describe the 50,000-transition training budget as total target access. Figure 3 in Appendix D reports realized access by environment and method. Learning curves use cumulative target training plus validation interactions on the horizontal axis, which permits the registered method-specific checkpoint grids without treating simulator steps as target access.

Table 2: Registered per-seed environment access. Episode lengths determine the realized validation and test transition counts, which are recorded in the interaction ledger.

Method	Simulator train	Target train	Validation	Final test
Calibration-routed MC-WM	50,000	50,000	8×5 episodes	20 episodes
Residual MC-WM core	50,000	50,000	8×5 episodes	20 episodes
Direct target MBPO / target SAC	0	50,000	8×5 episodes	20 episodes
DARC	50,000	50,000	8×5 episodes	20 episodes
Raw simulator SAC	50,000	0	8×5 episodes	20 episodes

4.4 REGISTERED COUNTERFACTUAL AND ABLATIONS

The exact routing study reuses the 30 final-candidate cells and adds 60 forced controls. All arms fit both candidates with equal non-routing arguments and frozen pre-deployment diagnostics; the automatic arm must exactly reproduce the metric stream and sanitized summary of its selected forced arm. The registered contrast is automatic routing minus the unselected alternative. Separate $3 \times 8 \times 10 = 240$ predecessor-mechanism and $3 \times 2 \times 10 = 60$ fixed-route sparse-versus-dense matrices test component and residual-backend claims. Appendices C.1, C.2, and C.3 give the respective interventions and matching rules.

4.5 ENDPOINTS, DIAGNOSTICS, AND STATISTICS

The statistical unit is one independently trained seed. For each environment and method, we report undiscounted episode return and counted violations, aggregated as the mean and sample standard deviation (SD) of seed-level 20-episode test means. Pre-registered paired comparisons subtract outcomes within the same environment and training seed and use a 20,000-resample percentile-bootstrap 95% confidence interval (CI) for the cross-seed mean. The main study uses candidate minus control. The routing study reports both candidate minus each forced arm and, as its causal contrast,

Table 3: Held-out target-domain outcomes over ten independently trained seeds (mean $\pm$ sample SD). Higher return and fewer violations are better. Test seeds are fixed across methods; target training, validation, and test access are reported separately in Appendix D.

Method	Ant actuator		HalfCheetah gravity		Walker friction	
	Return $\uparrow$	Violations $\downarrow$	Return $\uparrow$	Violations $\downarrow$	Return $\uparrow$	Violations $\downarrow$
Calibration-routed MC-WM	4084.9 $\pm$ 2105.9	1.39 $\pm$ 1.19	4954.8 $\pm$ 488.6	0.20 $\pm$ 0.40	394.6 $\pm$ 68.4	0.31 $\pm$ 0.09
MC-WM core	1361.7 $\pm$ 527.6	1.38 $\pm$ 1.48	4609.9 $\pm$ 602.0	2.48 $\pm$ 5.31	434.4 $\pm$ 247.8	24.98 $\pm$ 59.24
Direct target MBPO	4011.5 $\pm$ 3346.2	1.26 $\pm$ 1.32	4827.5 $\pm$ 1043.2	0.66 $\pm$ 0.89	350.7 $\pm$ 100.7	23.59 $\pm$ 21.45
Target SAC	2159.2 $\pm$ 461.1	3.95 $\pm$ 1.63	3770.3 $\pm$ 644.4	9.52 $\pm$ 10.24	261.2 $\pm$ 98.8	7.16 $\pm$ 8.19
DARC	1862.2 $\pm$ 534.7	1.80 $\pm$ 2.38	1268.6 $\pm$ 528.1	44.23 $\pm$ 26.03	252.5 $\pm$ 138.0	17.26 $\pm$ 19.03
Raw simulator SAC	1588.6 $\pm$ 416.2	0.41 $\pm$ 0.35	1088.8 $\pm$ 834.0	9.86 $\pm$ 5.26	408.8 $\pm$ 279.9	33.12 $\pm$ 68.54

automatic routing minus the forced alternative selected per seed. The mechanism matrix uses ablation minus predecessor, and the sparse-versus-dense study uses dense minus sparse. We label a direction only when its CI excludes zero, report every favorable, unfavorable, and unresolved comparison, and make no unregistered family-wise claim. Returns are not pooled across environments because their reward scales differ.

Each final-candidate run additionally reports the selected world-model family, both calibration risks, active and deployed sparse-support sizes, cross-seed support Jaccard overlap, confidence weights, and predicate rejection rate. The confidence diagnostic uses the first 1,000 stored target-replay transitions at the final model fit, and rejection rates aggregate imagined-rollout checks. The sparse-versus-dense study also reports calibration transition error and trainable state-residual parameters for both backends. These diagnostics test route consistency, quantify model fit and capacity, and distinguish inspection of one fitted symbolic correction from stability of that correction across seeds. No support-stability claim is inferred from equation export alone; Appendix B gives the fixed display rule.

The four matrices contain 540 unique reported cells from 541 completed executions. Aggregation rejects provenance, seed, budget, evaluation, or artifact drift, and generates every table and figure from validated artifacts. Appendix G gives the matrix ledger, replacement execution, frozen identities, and generation command.

5 RESULTS

5.1 HELD-OUT MAIN COMPARISON

Table 3 reports the one-shot held-out outcomes for the six methods. Figure 2 in Appendix D shows validation trajectories against cumulative target access. These trajectories can trigger candidate checkpoint restoration; held-out test episodes remain disjoint from that decision.

Relative to target SAC, calibration-routed MC-WM increases return on Ant by 1925.7 [715.6, 3322.3], on HalfCheetah gravity by 1184.5 [724.5, 1752.7], and on Walker friction by 133.4 [67.6, 197.9]. The paired violation differences are -2.56 [-3.74 , -1.45], -9.32 [-15.79 , -3.71], and -6.85 [-11.84 , -2.25], respectively; all three favor MC-WM. Candidate return also exceeds DARC in every environment. DARC violation differences favor MC-WM on gravity and Walker and are unresolved on Ant.

Direct target MBPO is the strongest model-based comparator. Its return interval against the candidate includes zero in all three environments: Ant 73.4 [-1947.6 , 1569.0], gravity 127.3 [-398.3 , 731.5], and Walker 43.9 [-32.2 , 120.4]. On Walker, however, MC-WM reduces violations by -23.29 [-36.18 , -11.21]. Compared with fixed residual MC-WM core, the candidate improves Ant return by 2723.2 [1502.9, 4056.3]; gravity and Walker return differences are unresolved, while Walker violations favor the candidate. Against raw simulator SAC, the candidate improves Ant and gravity return but incurs more Ant violations; Walker return is unresolved and Walker violations favor the candidate. These contrasts rule out a claim of dominance on every endpoint.

5.2 EXACT ROUTING COUNTERFACTUAL

The calibration criterion selects the direct target family in all ten Ant and all ten gravity runs, and the residual family in all ten Walker runs (Table 4). Every automatic run exactly matches its selected

Table 4: Exact route intervention over ten paired seeds. Selected R/D is the automatic residual/direct count; exact match compares the complete metric stream with the selected forced arm. Differences are automatic routing minus the unselected forced alternative, so positive return and negative violations favor the selector.

Environment	Selected R/D	Exact match	Return difference (95% CI)	Violation difference (95% CI)
Ant actuator	0/10	10/10	1529.9 [426.9, 2833.1]	0.72 [-0.05, 1.60]
HalfCheetah gravity	0/10	10/10	188.0 [-178.7, 591.7]	0.08 [-0.11, 0.37]
Walker friction	10/0	10/10	-21.2 [-56.0, 8.7]	-0.04 [-0.19, 0.06]

forced arm, establishing that the override itself does not perturb the selected training trajectory. The causal contrast therefore compares automatic routing with the unselected forced alternative under matched random streams.

The automatic-minus-alternative return differences are 1529.9 [426.9, 2833.1] on Ant, 188.0 [-178.7, 591.7] on gravity, and -21.2 [-56.0, 8.7] on Walker. The corresponding violation differences are 0.72 [-0.05, 1.60], 0.08 [-0.11, 0.37], and -0.04 [-0.19, 0.06]. Ant supplies a resolved route-return benefit; gravity and Walker return effects and all route-level violation effects are unresolved.

5.3 MECHANISMS, CAPACITY, AND INSPECTABILITY

The studies in Appendix C distinguish predecessor mechanisms from the final selector. The predecessor matrix supports validation checkpoint restoration on Ant and Walker, avoiding a fixed residual route on Ant, and a return-violation tradeoff from the violation cost; separate confidence and predicate removals remain unresolved. The fixed-route sparse comparison resolves no control endpoint, but uses fewer state-residual parameters in every environment while calibration risk favors different backends by environment. Auditability records show low cross-seed support overlap despite per-run equation export. Appendix Tables 6, 7, and 8 report the complete contrasts and diagnostics. The exported sparse-coordinate corrections in Equations (14), (15), and (16) make this per-run inspectability concrete; they are not governing laws shared across seeds.

6 DISCUSSION

The strongest result is model-family choice rather than residual modeling alone. Calibration risk routes Ant and gravity to direct target models and Walker to residual models. Exact selected-arm equality rules out an independent override effect; Ant shows a return benefit with no resolved route harm elsewhere. The main matrix also favors MC-WM over target SAC and matched DARC, but return relative to direct target MBPO remains unresolved. The supported claim is therefore calibrated choice under these shifts, not universal routing or model-based dominance.

The predecessor matrix supports checkpoint restoration, Ant fixed-route avoidance, and a violation-cost tradeoff, but not separate universal effects of confidence or predicates. Sparse residuals use fewer parameters and export per-run equations, yet show no endpoint advantage, mixed calibration error, and low cross-seed support overlap. Inspectability holds; stable structural discovery and sparse control superiority do not.

6.1 LIMITATIONS

The evidence covers three controlled MuJoCo shifts, not physical sim-to-real transfer or within-task route switching. DARC is the only external adaptation method under the identical access contract. Every adaptive method receives 50,000 target training interactions plus validation and test access, and the candidate may use validation for actor restoration; reduced total access and one-shot adaptation are not established. Confidence and predicates are empirical weights, not uncertainty bounds or safety guarantees. The Ant observation omits absolute planar position, although its radial reward and wall termination depend on that position. Its observation-space MDP is therefore an approximation: identical model inputs can have different rewards or wall outcomes, limiting the reward and termination heads. Broader shifts, matched adaptation baselines, and physical evaluation remain necessary.

ARTIFICIAL INTELLIGENCE USE STATEMENT

Artificial intelligence (AI) assistance through OpenAI Codex was used for literature and citation checks, feedback on method and experiment design, implementation audits and code editing, experiment orchestration, statistical aggregation, result interpretation, and manuscript drafting. The authors selected the research questions and frozen protocol, approved the changes, and reviewed the source metadata, mathematical statements, code, and reported artifacts. No large language model was called inside the reported training or evaluation runs, generated an experimental outcome, or selected which completed test results to report. The authors take responsibility for the final text, claims, code, and artifacts.

ETHICS STATEMENT

The study uses simulated control environments and does not involve human participants, personal data, animals, or physical-robot deployment. Confidence and predicate weights are empirical model-learning mechanisms and are not a formal safety guarantee.

REPRODUCIBILITY STATEMENT

Section 4 defines the environment shifts, comparators, interaction accounting, seeds, and statistical procedure. Appendix E lists the main hyperparameters, and Appendix G identifies the frozen source and the single command that regenerates the validated submission outputs. The anonymous supplemental archive contains source code, environment definitions, per-seed endpoint values, validation curves, target-access ledgers, symbolic supports, figure source data, sanitized per-run manifests and metrics, and the regeneration scripts. The manifests preserve experimental provenance but replace the machine-local run directory and Git revision; checkpoints are excluded because no reported result requires them. A public repository and immutable Git revision will be disclosed after double-blind review.

REFERENCES

- Joshua Achiam, David Held, Aviv Tamar, and Pieter Abbeel. Constrained policy optimization. *arXiv preprint arXiv:1705.10528*, 2017. URL <https://arxiv.org/abs/1705.10528v1>.
- Usman Anwar, Shehryar Malik, Alireza Aghasi, and Ali Ahmed. Inverse constrained reinforcement learning. *arXiv preprint arXiv:2011.09999*, 2020. URL <https://arxiv.org/abs/2011.09999v3>.
- Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data: Sparse identification of nonlinear dynamical systems. *arXiv preprint arXiv:1509.03580*, 2015. URL <https://arxiv.org/abs/1509.03580v1>.
- Xinyue Chen, Che Wang, Zijian Zhou, and Keith Ross. Randomized ensembled double Q-learning: Learning fast without a model. *arXiv preprint arXiv:2101.05982*, 2021. URL <https://arxiv.org/abs/2101.05982v2>.
- Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. *arXiv preprint arXiv:1805.12114*, 2018. URL <https://arxiv.org/abs/1805.12114v2>.
- Benjamin Eysenbach, Shreyas Chaudhari, Swapnil Asawa, Sergey Levine, and Ruslan Salakhutdinov. Off-dynamics reinforcement learning: Training for transfer with domain classifiers. *arXiv preprint arXiv:2006.13916*, 2020. URL <https://arxiv.org/abs/2006.13916v2>.
- Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. *arXiv preprint arXiv:1801.01290*, 2018. URL <https://arxiv.org/abs/1801.01290v2>.

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- Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. *arXiv preprint arXiv:1912.01603*, 2019. URL <https://arxiv.org/abs/1912.01603v3>.
- Nicklas Hansen, Xiaolong Wang, and Hao Su. Temporal difference learning for model predictive control. *arXiv preprint arXiv:2203.04955*, 2022. URL <https://arxiv.org/abs/2203.04955v2>.
- Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. *arXiv preprint arXiv:1906.08253*, 2019. URL <https://arxiv.org/abs/1906.08253v3>.
- Ashish Kumar, Zipeng Fu, Deepak Pathak, and Jitendra Malik. RMA: Rapid motor adaptation for legged robots. *arXiv preprint arXiv:2107.04034*, 2021. URL <https://arxiv.org/abs/2107.04034v1>.
- JB Lanier, Kyungmin Kim, Armin Karamzade, Yifei Liu, Ankita Sinha, Kat He, Davide Corsi, and Roy Fox. Adapting world models with latent-state dynamics residuals. *arXiv preprint arXiv:2504.02252*, 2025. URL <https://arxiv.org/abs/2504.02252v1>.
- Haoyi Niu, Tianying Ji, Bingqi Liu, Haocheng Zhao, Xiangyu Zhu, Jianying Zheng, Pengfei Huang, Guyue Zhou, Jianming Hu, and Xianyuan Zhan. H2O+: An improved framework for hybrid offline-and-online RL with dynamics gaps. *arXiv preprint arXiv:2309.12716*, 2023. URL <https://arxiv.org/abs/2309.12716v2>.
- Xue Bin Peng, Marcin Andrychowicz, Wojciech Zaremba, and Pieter Abbeel. Sim-to-real transfer of robotic control with dynamics randomization. *arXiv preprint arXiv:1710.06537*, 2017. URL <https://arxiv.org/abs/1710.06537v3>.
- Stéphane Ross and J. Andrew Bagnell. Agnostic system identification for model-based reinforcement learning. *arXiv preprint arXiv:1203.1007*, 2012. URL <https://arxiv.org/abs/1203.1007v2>.
- Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. *arXiv preprint arXiv:1703.06907*, 2017. URL <https://arxiv.org/abs/1703.06907v1>.
- Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 5026–5033, 2012. doi: 10.1109/IROS.2012.6386109. URL <https://doi.org/10.1109/IROS.2012.6386109>.

A COMPLETE NOTATION LEDGER

Table 5 fixes one meaning for every nonstandard symbol used in the manuscript. Standard arithmetic operators retain their usual meaning. Indices are fixed as follows: t is interaction time, i is a data sample, i' is an auxiliary sample in a normalization, k is a joint prediction-output coordinate ($1 \leq k \leq n_s + 1$), j is a policy critic ($1 \leq j \leq 3$), m is a zero-based state coordinate ($0 \leq m < n_s$), n is a zero-based action coordinate ($0 \leq n < n_a$), u is a validation-checkpoint index, m is a candidate world-model family, and ℓ is a domain. Generic s , a , and x omit the interaction index; s' denotes a successor state. Hats denote predictions, and the superscript (R) restricts a prediction to the residual candidate. Parameters retain the same roles: ϕ for the policy, ψ for the neural state remainder, ζ for the reward correction, ν for the termination head, ξ for the confidence critic, and η_j for policy critic j ; $\bar{\xi}$ is the confidence target’s parameter vector. $\mathbb{R}$ denotes the real numbers, $\|\cdot\|_2$ the Euclidean norm, $\odot$ elementwise multiplication, $|\mathcal{I}_{\text{cal}}|$ set cardinality, $|\cdot|$ on scalars absolute value, and $\mathbb{I}[\cdot]$ the Boolean indicator.

Table 5: Complete mathematical notation.

Symbol	Unique meaning
$\mathcal{S}; \mathcal{A}$	state space; action space
$\mathcal{M}_\ell$	domain MDP, with $\ell \in \{\text{S}, \text{T}\}$ denoting simulator or target
$P_\ell; \rho_\ell$	transition kernel; expected one-step reward in domain ℓ
$\gamma; p_0$	policy-return discount; initial-state distribution
π_ϕ	policy with parameter vector ϕ
$J_{\mathcal{M}}(\pi)$	discounted return of policy π in MDP $\mathcal{M}$
$\mathbb{E}$	expectation under the subscripted process
$s_t; a_t; r_t; \kappa_t$	observed state; action; observed reward; observed termination indicator at time t
$s^{(m)}; a^{(n)}$	coordinate m of a generic state; coordinate n of a generic action
$x_t = [s_t; a_t]$	concatenated state-action input
$n_s; n_a; p$	state dimension; action dimension; deployed feature count
$\mathcal{D}_\text{S}; \mathcal{D}_\text{T}; \mathcal{D}_\text{T}^0$	simulator training dataset; target training dataset; initial target dataset available before online interaction
$\mathcal{I}_\text{fit}; \mathcal{I}_\text{sel}; \mathcal{I}_\text{cal}$	initial target fit; feature-selection and early-stopping; calibration index set
$\widehat{M}_\text{S}; \widehat{M}_\text{R}; \widehat{M}_\text{D}; \widehat{M}_\text{T}$	frozen simulator model; residual candidate; direct target candidate; deployed target model
$z_i; \widehat{z}_i^{(\mathbf{m})}$	joint state-increment and reward target; its prediction from candidate family $\mathbf{m}$
ς_k	fit-partition scale for joint-target coordinate k
$\mathbf{m}; \mathbf{m}^*$	candidate world-model family; deployed family
$\mathcal{R}_\mathbf{m}$	standardized calibration transition risk for family $\mathbf{m}$
$\widehat{u}_\text{S}(x); \widehat{\rho}_\text{S}(x)$	simulator state-increment; reward prediction
$\widehat{\Delta}_\text{s}(x); \widehat{\Delta}_{r,\zeta}(x)$	target state; reward correction, with reward parameters ζ
$\widehat{\Delta}_{\text{s,sp}}^{(m)}(s, a)$	coordinate m of the reported sparse state-correction branch
$\widehat{p}_\nu(s, a, s')$	termination probability with parameters ν
$\widehat{s}_{t+1}; \widehat{r}_t; \widehat{\kappa}_t$	deployed-model state; reward; termination prediction
$y_i^{\Delta s}$	target state-correction regression target for sample i
$\mathcal{F}; \Theta(x); \widetilde{\Theta}(x)$	candidate feature set with fixed evaluation order; its unscaled evaluation vector; scaled deployed feature vector
$B; \mu$	sparse state-correction coefficient matrix; binary feature-support mask
g_ψ	masked neural state remainder with parameters ψ
$f; \mathcal{L}_\text{sel}; \epsilon_\text{feat}$	candidate feature; selection loss; required selection improvement
$\omega_i; d_i; h_d$	policy-visitation refit weight; root mean squared standardized Euclidean distance to eight neighbors; positive distance bandwidth
$e_i; \tau_e; c_i$	state prediction MSE; running median-error scale; one-step confidence target
$\sigma_\text{ens}(s, a)$	coordinatewise state-increment and reward standard deviation from the route-specific uncertainty ensemble
$Q_\xi^\Delta; Q_\xi^\Delta$	log-confidence critic; Polyak-averaged target critic
a_{i+1}^det	deterministic action of the current policy at sample i 's successor
$\gamma_\Delta; q_i^{\text{tgt}}$	confidence discount; confidence regression target
w_i^Δ	imagined-transition confidence weight
$\mathcal{C}; \chi_i$	deterministic predicate set; binary predicate-pass indicator
w_i	final policy-critic loss weight for transition i
$\mathcal{I}; b$	policy-optimization minibatch index set; its size
$Q_{\eta_j}; y_i^Q$	policy critic j with parameters η_j ; standard SAC target
$\mathcal{L}_Q; \lambda_\text{ens}$	weighted policy-critic loss; critic-disagreement coefficient
$\nu_i; \lambda_v(t); r_i^{\text{train}}$	binary violation indicator; online violation cost; reward consumed by policy training

Symbol	Unique meaning
ϱ_{cal}	initial simulator-to-residual state RMSE ratio on the calibration partition
$\overline{G}_u; \overline{V}_u$	mean undiscounted validation return; mean validation violations per episode at checkpoint u
$\overline{G}_{\text{peak}}; \overline{G}_{\text{final}}$	peak; final-checkpoint mean undiscounted validation return
$K; N_{\text{on}}$	member count in either probabilistic world-model ensemble; online target-training budget
$q_z; v_x$	HalfCheetah root vertical generalized coordinate; Ant forward velocity

B AUDITABILITY DIAGNOSTICS

Table 6 reports diagnostics from the ten complete MC-WM runs in each environment. Support and state-error columns describe the residual candidate even when the direct family is deployed: the initial fit on Ant and gravity, and the final residual refit on Walker. SINDy MSE and additive MSE are unstandardized state-correction errors on that fit’s calibration partition; the error 95th percentile is the Euclidean norm of the additive state error, not the standardized joint routing risk in Equation (3). The support Jaccard statistic compares active coordinate-feature pairs between every pair of training seeds. Low overlap is evidence against cross-seed structural stability; it does not prevent a run’s locked equation and coefficients from being inspected. Confidence is evaluated on the fixed target-replay prefix defined in Section 4; the rejection rate aggregates every predicate check made during imagined rollouts.

Table 6: Per-run model diagnostics aggregated across training seeds. Parentheses contain cross-seed SD unless a range is shown.

Environment	Active coeff.	Support overlap	SINDy MSE	Additive MSE	Error 95th pct.	Mean confidence	Reject rate
Ant, actuator failure	228.9 (50.5)	0.24 [0.19, 0.30]	0.34 (0.04)	0.21 (0.06)	4.32 (0.73)	0.23 (0.08)	0.0% (0.0)
HalfCheetah, soft gravity	271.2 (58.9)	0.26 [0.16, 0.37]	0.98 (0.94)	0.40 (0.05)	5.24 (0.32)	0.27 (0.02)	35.1% (8.4)
Walker2d, friction	210.5 (30.2)	0.20 [0.15, 0.26]	2.53 (1.07)	0.56 (0.20)	6.22 (1.01)	0.26 (0.04)	55.9% (4.5)

Equations (14), (15), and (16) give the complete, untruncated representative coordinate corrections selected by the fixed rule below. They report only $B\hat{\Theta}$, not the neural remainder in Equation (8):

Ant, actuator failure (support-medoid seed 952414860).

$$\hat{\Delta}_{s,\text{sp}}^{(9)}(s, a) = -0.138 a^{(6)} \quad (14)$$

HalfCheetah, soft gravity (support-medoid seed 952414860).

$$\hat{\Delta}_{s,\text{sp}}^{(7)}(s, a) = -0.128 s^{(0)} s^{(1)} \quad (15)$$

Walker2d, friction (support-medoid seed 295539546).

$$\begin{aligned} \hat{\Delta}_{s,\text{sp}}^{(9)}(s, a) = & -8.6 + 7.04 s^{(0)} - 9.02 s^{(1)} + 2.04 s^{(2)} \\ & + 0.943 s^{(3)} - 1.07 s^{(5)} + 4.2 s^{(9)} - 0.0474 s^{(15)} \\ & - 0.0382 s^{(16)} + 3.74 s^{(0)} s^{(1)} - 0.143 s^{(0)} s^{(2)} + 0.399 s^{(0)} s^{(5)} \\ & - 2.22 s^{(0)} s^{(9)} + 0.033 s^{(0)} s^{(15)} + 0.0574 s^{(0)} s^{(16)} - 0.44 s^{(1)} s^{(3)} s^{(6)} \\ & + 1.65 s^{(0)} s^{(3)} s^{(6)} \end{aligned} \quad (16)$$

The supplemental auditability directory contains `full_symbolic_support.json`, which stores every active term and coefficient for all thirty final-candidate runs.

For display only, the representative seed maximizes mean support Jaccard to the other nine seeds. Within that run, the fixed rule prefers nonempty coordinates with at most eight terms, then orders them by cross-seed coordinate support Jaccard, fewer terms, and output label. This rule does not affect model fitting, route selection, or endpoint analysis.

C REGISTERED MECHANISM AND RESIDUAL STUDIES

C.1 EXACT ROUTING COUNTERFACTUAL

On the fresh confirmation seeds, all three arms execute the same residual and direct-target candidate fitting code with matched data, initialization, and isolated component random streams. Direct-candidate fitting saves and restores the global NumPy and PyTorch random states, including visible accelerators; its dedicated stream is derived deterministically from the training seed. The automatic arm deploys the lower-risk family; the controls force residual or direct deployment only after both risks have been recorded. Violation costs, trust filters, actor updates, termination treatment, and checkpoint restoration are identical. The analyzer requires equal non-routing arguments and diagnostics frozen before deployment; mutable online residual-refit diagnostics are excluded. For each seed, the automatic arm must also have a byte-identical metric stream and equal summary, after removing two route-label fields, to the forced arm for its selected family.

C.2 PREDECESSOR MECHANISM MATRIX

A source-identical $3 \times 8 \times 10$ matrix starts from the frozen gap-routed predecessor and removes model-family routing, confidence weighting, deterministic predicates, the violation cost, validation checkpoint restoration, or the sparse state branch. A seventh control removes both routing and the sparse branch, allowing a fixed-residual sparse contrast. This matrix tests the predecessor; it is not relabelled as an ablation of the final calibration-risk selector.

Table 7: Paired component matrix for the frozen gap-routed predecessor. Differences are ablation minus predecessor; negative return therefore favors the predecessor, while negative violations favor the ablation. This matrix is not a source-identical ablation of the final calibration-risk selector.

Environment	Ablation	Return difference (95% CI)	Violation difference (95% CI)
Ant actuator	No checkpoint selection	-428.8 [-650.1, -212.8]	-0.01 [-0.32, 0.33]
Ant actuator	No confidence weighting	593.8 [-2024.7, 3881.6]	2.28 [-0.38, 6.29]
Ant actuator	Fixed residual route	-2231.8 [-4461.5, -871.3]	-0.50 [-1.18, 0.17]
Ant actuator	Fixed residual route, neural only	-2302.4 [-4409.3, -901.8]	-0.57 [-1.23, 0.02]
Ant actuator	No predicate filtering	-162.2 [-2178.5, 1960.5]	1.43 [-0.72, 4.96]
Ant actuator	No sparse branch	0.0 [0.0, 0.0]	0.00 [0.00, 0.00]
Ant actuator	No violation cost	1560.0 [-960.7, 4793.5]	17.02 [0.83, 46.07]
HalfCheetah gravity	No checkpoint selection	57.1 [0.0, 171.3]	0.17 [0.00, 0.52]
HalfCheetah gravity	No confidence weighting	-150.6 [-809.6, 685.1]	-0.15 [-0.93, 0.52]
HalfCheetah gravity	Fixed residual route	-202.1 [-477.8, 0.0]	-0.25 [-0.62, 0.00]
HalfCheetah gravity	Fixed residual route, neural only	-232.2 [-703.1, 287.4]	0.08 [-0.77, 0.77]
HalfCheetah gravity	No predicate filtering	-332.9 [-888.4, 230.9]	0.17 [-0.79, 1.09]
HalfCheetah gravity	No sparse branch	-175.0 [-599.0, 316.5]	-0.06 [-0.88, 0.64]
HalfCheetah gravity	No violation cost	89.2 [-197.2, 411.0]	2.55 [-0.17, 6.33]
Walker friction	No checkpoint selection	-84.2 [-158.8, -24.9]	-0.78 [-2.32, 0.01]
Walker friction	No confidence weighting	-11.3 [-87.1, 71.5]	-0.76 [-2.30, 0.03]
Walker friction	Fixed residual route	0.0 [0.0, 0.0]	0.00 [0.00, 0.00]
Walker friction	Fixed residual route, neural only	-26.6 [-188.1, 150.9]	-0.68 [-2.24, 0.13]
Walker friction	No predicate filtering	-6.2 [-173.7, 167.1]	-0.69 [-2.15, 0.07]
Walker friction	No sparse branch	-37.9 [-193.7, 135.6]	-0.69 [-2.24, 0.12]
Walker friction	No violation cost	58.6 [6.7, 118.9]	0.12 [0.03, 0.24]

C.3 SPARSE VERSUS DENSE STATE RESIDUAL

A fresh $3 \times 2 \times 10$ matrix compares the sparse additive residual plus masked neural remainder with a two-hidden-layer dense neural residual. Both arms force residual deployment and match target access, source-model data, reward and termination heads, trust filters, policy updates, and evaluation. The dense arm uses standardized raw state-action inputs and performs no feature discovery or symbolic regression. Primary endpoints are return and violations; calibration transition error and state-residual parameter count are secondary.

Table 8: Paired dense-minus-sparse residual comparison under fixed residual deployment. Calibration risk is standardized transition error, and state parameters count trainable parameters in the state-correction branch.

Environment	Return difference (95% CI)	Violation difference (95% CI)	Calibration-risk difference (95% CI)	State-parameter difference (95% CI)	Classification
Ant actuator	29.7 [-546.1, 652.8]	-0.04 [-0.81, 0.72]	0.629 [0.485, 0.796]	1414 [1396, 1431]	Unresolved
HalfCheetah gravity	156.3 [-192.8, 506.9]	0.66 [-0.24, 2.18]	-0.103 [-0.170, -0.047]	1347 [1308, 1401]	Unresolved
Walker friction	45.4 [-56.3, 147.4]	0.33 [-0.08, 0.99]	-0.007 [-0.010, -0.004]	1386 [1337, 1440]	Unresolved

D TARGET-ACCESS LEDGER AND LEARNING CURVES

Figure 3 separates target training, validation, and final test interactions. Figure 2 plots validation return against cumulative target access through that checkpoint; its horizontal axis therefore includes validation interactions. The registered candidate does not use validation outcomes for gradient updates, but it can use them to select the actor evaluated at the final test.

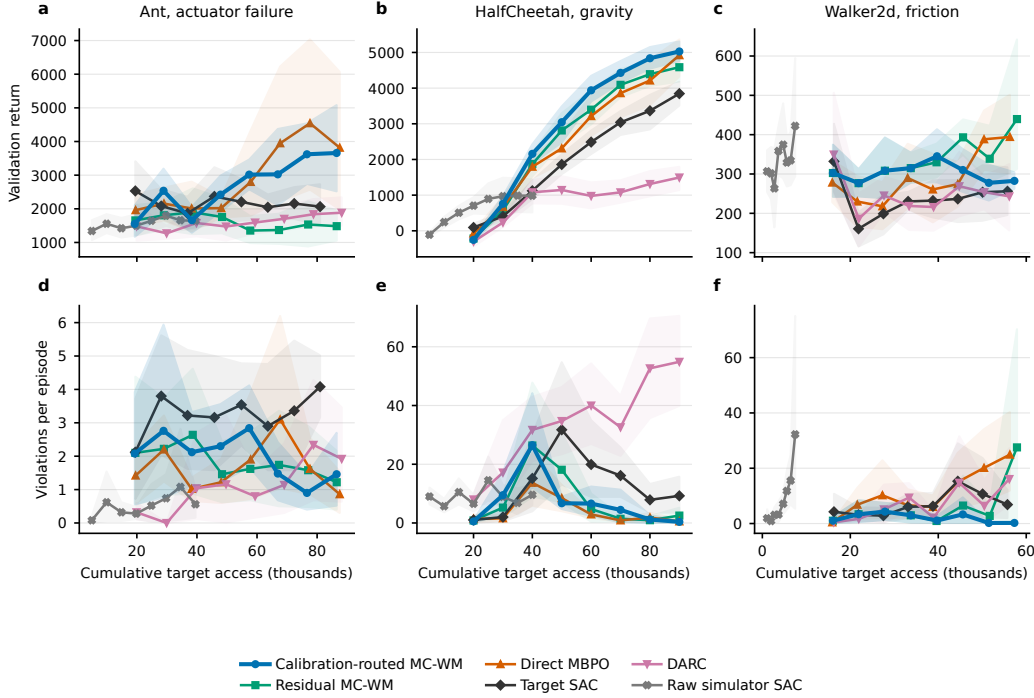


Figure 2: Validation return and violations against cumulative target access. Lines are ten-seed means and bands are 20,000-resample percentile-bootstrap 95% confidence intervals for the cross-seed mean. The horizontal coordinate includes target training plus cumulative validation interactions. Validation outcomes do not enter gradient updates, but the registered candidate may use its validation trajectory for checkpoint restoration.

E IMPLEMENTATION DETAILS

Table 9 lists the fixed calibration-routed candidate settings.

E.1 RESIDUAL FITTING DETAILS

STLSQ alternates ridge regression and coefficient thresholding, both set to 0.05. An additional deterministic orthogonal feature f is accepted only when

$$\mathcal{L}_{\text{sel}}(\mathcal{F} \cup \{f\}) < \mathcal{L}_{\text{sel}}(\mathcal{F}) - \epsilon_{\text{feat}}, \quad \epsilon_{\text{feat}} = 10^{-5}, \quad (17)$$

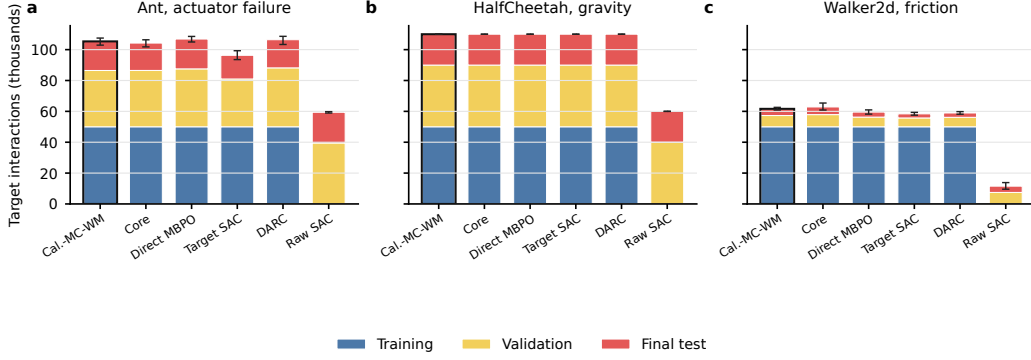


Figure 3: Actual target access by method and environment. Bars show cross-seed means for training, validation, and final-test transitions; error bars show the 20,000-resample percentile-bootstrap 95% CI of mean total access.

Table 9: Fixed MC-WM hyperparameters used by every calibration-routed candidate run.

Component	Parameter	Value
Simulator ensemble	members / hidden layers / width	5/3/200
Simulator ensemble	batch size / learning rate	256/ 10^{-3}
Residual split	fit / selection / calibration	0.8/0.1/0.1
Model-family route	metric / tie rule	standardized joint RMSE / residual
Sparse regression	ridge / threshold	0.05/0.05
Feature discovery	budget / minimum gain	100/ 10^{-5}
Neural remainder	hidden width / learning rate	32/ 10^{-3}
Reward and termination heads	hidden width	64
Online deployed-model refit	frequency / maximum rows	1,000/50,000
Imagined data	horizon / batch / frequency	1/400/250
Target / imagined replay capacity	transitions	100,000/50,000
Confidence critic	hidden width / γ_{Δ}	128/0.5
Policy learner	critics / hidden width	3/256
Policy learner	γ / learning rate	0.99/(3×10^{-4})
Policy learner	pessimism / disagreement coefficient	-2.0/0.01
Online exploration	uniform-action cutoff	$t < 2,000$
Actor route	mixed-update calibration band	[2.0, 4.5)
Mixed actor	synthetic trust power / update interval	0.5/4 critic updates
Standard actor	update interval	2 critic updates
Violation cost	start / final value	0.5 N_{on} /10
Validation actor selection	violation budget / return retention	1.0/0.8

where an admitted f is appended to the fixed feature order, and $\mathcal{L}_{\text{sel}}$ is the selection-partition MSE after refitting on $\mathcal{I}_{\text{fit}}$. Before this gate, an orthogonal candidate is discarded when its largest estimated absolute coefficient exceeds 5. This is a local stability screen, not a bound on the complete coefficient matrix. The feature budget is $p \leq 100$; Equation (17) is the only selection-loss acceptance rule.

After discovery, the feature order and support mask μ are locked. Residual-route refits update B , g_{ψ} , and the reward and termination heads without adding features. The latest 10,000 state-action pairs form the policy reference bank; the distance calculation uses a reproducible 8,000-row subsample when the bank is larger. For refit sample i , let d_i be the square root of the mean squared standardized Euclidean distance to its eight nearest retained reference points (with 10^{-8} added inside the square root), and let h_d be the median of these distances plus 10^{-6} . Its refit weight is

$$\omega_i = 0.1 + 0.9 \frac{\exp(-d_i/h_d)}{\max_{i'} \exp(-d_{i'}/h_d)}. \quad (18)$$

The index i' ranges over the refit samples. Equation (18) weights the sparse and neural state losses; it is not an importance ratio or a value-error bound.

A separate 64-unit ReLU classifier estimates target termination probability $\hat{p}_\nu(s, a, s')$. The deployed termination prediction is

$$\hat{\kappa}_t = \mathbb{I}[\hat{p}_\nu(s_t, a_t, \hat{s}_{t+1}) \geq 0.5], \quad (19)$$

where ν denotes classifier parameters and $\mathbb{I}[\cdot]$ is the indicator function. The initially fitted head is carried onto either route; residual-route refits update it. Equation (19) is consumed only when the observed target-replay termination rate exceeds 2%, avoiding an effectively unidentified rare-event classifier.

E.2 ACTOR ROUTING, CHECKPOINT SELECTION, AND COMPLETE PROCEDURE

The initial state-only calibration ratio ϱ_{cal} divides the frozen simulator state RMSE by the residual candidate state RMSE on $\mathcal{I}_{\text{cal}}$. When $2 \leq \varrho_{\text{cal}} < 4.5$, actor updates use equal numbers of target and imagined states; imagined actor samples carry weight $(w_i)^{1/2}$, and the actor updates after every four critic updates. Outside this band, target and imagined replay use the standard separate SAC updates.

At each 5,000-step checkpoint, five fixed validation episodes yield mean undiscounted return $\bar{G}_u$ and mean violations per episode $\bar{V}_u$. MC-WM tracks the highest-return actor satisfying $\bar{V}_u \leq 1$. Before the one-time test, that actor replaces the final actor only if either (i) final validation return is below the peak by more than $0.2 \max(|\bar{G}_{\text{peak}}|, 1)$ and the tracked actor has higher return, or (ii) final validation violations exceed one and the tracked return is at least the final return minus $0.2 \max(|\bar{G}_{\text{final}}|, 1)$. Here $\bar{G}_{\text{peak}} = \max_u \bar{G}_u$ is the maximum over all validation checkpoints, including ineligible ones, and $\bar{G}_{\text{final}}$ is the final-checkpoint mean. The critics and world model are not selected. Final testing uses 20 fixed seeds disjoint from training and validation, exactly once after any actor restoration.

Algorithm 1 Calibration-routed MC-WM training and held-out testing

Require: simulator data $\mathcal{D}_S$, initial target data $\mathcal{D}_T^0$, online target budget N_{on}

- 1: Fit and freeze the simulator ensemble $\widehat{M}_S$
 - 2: Split $\mathcal{D}_T^0$ into $\mathcal{I}_{\text{fit}}$, $\mathcal{I}_{\text{sel}}$, and $\mathcal{I}_{\text{cal}}$
 - 3: Fit $\widehat{M}_R$ and fit $\widehat{M}_D$ in an isolated component random stream
 - 4: Compute $\mathcal{R}_R$, $\mathcal{R}_D$, and ϱ_{cal} ; fix m^* and the actor-update route
 - 5: Lock residual feature order and support μ ; initialize target replay, imagined replay, π_ϕ , and Q_ξ^Δ
 - 6: Fit Q_ξ^Δ on initial target-replay prediction errors
 - 7: **for** $t = 1, \dots, N_{\text{on}}$ **do**
 - 8: Execute one target action and append the observed transition
 - 9: **if** t is a 1,000-step model-refit point **then**
 - 10: Warm-start refit only the deployed state-and-reward model
 - 11: Update Q_ξ^Δ from target-replay prediction errors
 - 12: **end if**
 - 13: **if** t is a 250-step imagination point **then**
 - 14: Generate one-step transitions from $\widehat{M}_T$
 - 15: Compute w_i^Δ , χ_i , w_i , and ι_i ; retain rows with $w_i > 0.01$
 - 16: **end if**
 - 17: Apply $\lambda_v(t)$ and update SAC using the fixed actor route
 - 18: **if** t is a 5,000-step validation point **then**
 - 19: Record $\bar{G}_u$ and $\bar{V}_u$; update the best eligible actor checkpoint
 - 20: **end if**
 - 21: **end for**
 - 22: Apply the registered trajectory restoration rule, then evaluate once on held-out test seeds
-

Direct target MBPO. This comparator fits the same five-member, three-layer, width-200 probabilistic ensemble directly to target transitions. It uses the same policy learner, one-step rollout size and frequency, and target and imagined replay capacities as MC-WM, but has no simulator pretraining, residual, confidence weight, or predicate. The ensemble fits for at most 100 epochs initially and is refit every 1,000 online steps for at most 20 epochs on at most 50,000 target transitions; both stages use a 10% internal holdout and early stopping.

Dense neural residual control. The frozen simulator receives a dense state-correction network with two width-64 ReLU hidden layers. Raw state-action inputs and state-correction targets are standardized from the initial fit partition; the disjoint selection partition supplies early stopping, and the calibration partition remains untouched. Adam uses learning rate 10^{-3} , weight decay 10^{-4} , and batch size 256. The reward and termination heads, policy-aware refit weights, trust filters, policy learner, and target-access schedule are identical to the sparse fixed-route arm. The dense arm performs no feature discovery and exports no symbolic coefficients.

Matched DARC implementation. The DARC comparator uses two binary domain classifiers. The first receives (s_t, a_t) ; the second has an additive state-action logit and a conditional next-state logit for (s_t, a_t, s_{t+1}) . Each hidden layer has width 32 and classifier inputs receive Gaussian noise with SD 1.0. Source rewards are adjusted by the difference between target-versus-source transition and state-action logits. Classifiers train from the same 50,000 source and 50,000 target training interactions available under the matched protocol. The 4,000-step correction warmup is disclosed because it differs from the original long-horizon schedule.

F EXACT ENVIRONMENT TRANSFORMATIONS

All modifications below are applied identically to training, validation, and test instances of the corresponding domain. In the HalfCheetah gravity environment, the simulator’s vertical gravity is set to -19.62 m s^{-2} and the target retains the MuJoCo default -9.81 m s^{-2} . Both members continue the episode when $q_z > 0.2$ and subtract $10[\max(0, q_z - 0.2)]^2$ from the standard HalfCheetah reward.

For Friction Walker, every sliding, torsional, and rolling geom-friction coefficient is multiplied by 0.3 in the simulator; the target retains the default coefficients. Both domains subtract $10[\max(0, s^{(0)} - 1.25)]^2$ from the standard Walker2d reward, where $s^{(0)}$ is the root-height observation.

For Ant actuator failure, target actions 4–7 are set to zero before the physics step; the simulator applies all eight actions. Both domains retain the standard Ant health termination, add a rear-wall termination when the world-frame base position falls below -3 along the first horizontal coordinate, and set the wall-step reward to zero. Away from the wall, the project-specific reward is planar Euclidean distance from the origin plus 1, minus 0.5 times the squared norm of the applied action. The soft cap further subtracts $[\max(0, |v_x| - 1.5)]^2$. Thus disabled target channels incur no control cost after they are zeroed. The reported Ant method has an empty deterministic predicate set; these environment rewards and terminations still apply.

G REPRODUCIBILITY AND FROZEN SOURCE

The final main matrix and exact forced-route controls record source-tree identity:

939a71c718b4f8d2296a41c5c43cd8323e84fdeedfd7ce374104d05ae420af9c0.

The predecessor mechanism matrix records:

66eb8a36ee482d3a22000b8302c91c8da4ac5914f59bbd14e9182f581f028a8e.

The sparse-versus-dense study records:

06e1d3eea255e83c7be769e364ebc59a2e63a4ee1feb473a81ecec24bf52853c.

Git revisions are withheld during double-blind review; later manuscript and packaging changes leave these experimental methods unchanged. One Ant forced-residual routing cell (seed 976484671) replaces task t88897 with t89031 after three pre-deployment diagnostics failed equality across nodes. The replacement used identical source, arguments, method, and seed on the candidate’s node and passed the exact gate. It was registered before completion or inspection of aggregated endpoints. The ledger contains 540 unique cells from 541 executions: 180 main comparisons, 60 additional route controls, 240 predecessor cells, and 60 sparse-versus-dense cells; reused candidates count once. The archive includes anonymized frozen sources, runnable code, sanitized per-run JavaScript Object Notation (JSON) and JSON Lines (JSONL) artifacts, figure source data, and per-run arguments, dependencies, seeds, metrics, and interaction ledgers. Run `experiments/`

`build_submission_outputs.sh` to regenerate assets from complete protocol-matched seed sets. No external language-model service is used by any reported matrix.